Newton's Second Law: Measuring the Mass of a System From a Graph

SYNOPSIS

Newton's second law says **a = F / M**. Written that way it is a claim about a straight line: if you hold the mass of a system fixed and increase the force driving it, the acceleration should rise *in proportion*, and the slope of acceleration against force should be **1 / M** — one over the total mass of everything that is moving. This lab puts both halves of that claim on one graph. A cart on a track is pulled by a hanging mass over a pulley. To raise the driving force without changing the mass being driven, you move mass from the cart to the hanger rather than adding any. Five runs of position-versus-time data give you five accelerations; plotting them against the five driving forces gives you a line whose slope is a mass you never put on a scale. Then you weigh the system and see whether the graph was telling the truth.

Due see the Module 3 Lab folder on Brightspace

Report Full lab report — use the Lab Template

Pairs with M3 N6

Data provided — this is a virtual lab

PERFORMANCE EXPECTATIONS what you should be able to do when this lab is written up

- **PE1** **Draw a free-body diagram for each body in a connected system** and mark the single acceleration they share, choosing a positive direction for each body separately along its own direction of motion.
- **PE2** **Write ΣF = ma for each body and combine the two equations** by eliminating the tension, arriving at $a = F_{g,H} / M_T$ without being handed it.
- **PE3** **Explain why mass is *transferred* and not added** — that is, why this procedure, and not the obvious one, is what makes force the single variable under test.
- **PE4** **Extract an acceleration from position-versus-time data** by fitting a parabola to the part of the run where the system is actually accelerating, and taking a = 2 × the quadratic coefficient.
- **PE5** **Interpret the slope of a graph as a physical quantity**, including its units — here, obtaining a mass from the slope of acceleration against force, together with the uncertainty that slope carries.
- **PE6** **Judge a verification against stated criteria**, deciding on the evidence whether Newton's second law was confirmed, and accounting for the difference between the mass from your graph and the mass from the scale.

Start from Note 6 — this lab is that note, measured

Note 6 is not background reading for this lab. It is the lab. The half Atwood machine you drew in class, the two free-body diagrams you filled in, and the derivation you worked through to $a = M_H g / (M_H + M_C)$ are the Introduction, the Experimental section and the Analysis plan of the report you are about to write. You are expected to bring that note into the report and build on it — not to cite it in passing, and not to re-derive it from scratch as though you had never seen it.

Where each part of Note 6 belongs in this report. The rubric awards points for these.

Report section	What Note 6 gives you
Introduction	The statement of the second law and what "verify" means here, plus **your own derivation** of $a = F_{g,H} / M_T$ from the two equations. Define every symbol. End with a hypothesis you can later confirm or reject.
Experimental	The labelled setup drawing and both free-body diagrams, redrawn by you. State the sign convention you chose for each body. Describe how the data were collected and — explicitly — **how the total system mass was held constant while the hanging mass increased, and why that matters.**
Analysis	The rearrangement $a = (1 / M_T) \times F_{g,H}$, read as the equation of a straight line: acceleration on the vertical axis, driving force on the horizontal, **slope = $1 / M_T$**. This is what tells you the graph can weigh the system.
Discussion	Three results from the note that your data should be checked against: the tension is **less than** the weight of the hanging mass; the acceleration is **less than** g; and the cart and the hanging mass share one acceleration even though they move in different directions.

Read or reprint the note here: Module 3 Note 6 — Verifying Newton's Second Law with a Half Atwood Machine.

The physics you are using

> **Sign convention here.** This system bends around a pulley, so a single pair of axes cannot serve both bodies. Choose a positive direction *for each body separately, along its own direction of motion*: for the cart, take the direction it travels along the track as positive; for the hanging mass, take **downward** as positive. Both bodies then have the same positive acceleration **a**, and the string carries the one tension **T** that connects them. This is the convention your note uses, and it is the reason the two equations combine as cleanly as they do.

The cart feels three forces, but only the tension acts along its direction of motion. Vertically the normal force and the weight cancel, so the cart's equation is just:

$$T = M_C a$$

Where this holds: a horizontal track with negligible friction, with the string pulling horizontally. If the track is not level, or if friction is not negligible, a term is missing from this equation — and you will see the evidence for that in your own graph.

The hanging mass feels its weight downward and the tension upward. Taking down as positive for this body:

$$F_{g,H} - T = M_H a \quad \text{where } F_{g,H} = M_H g$$

Note what this says before you solve it: because the right-hand side is positive, T must be *smaller* than $F_{g,H}$. The tension is not the weight of the hanging mass. If it were, the net force on the hanger would be zero and nothing would accelerate.

Eliminate T between the two and solve for the acceleration:

$$a = M_H\, g \,/\, (M_H + M_C) = F_{g,H} \,/\, M_T$$

Read the denominator carefully. It is the *total* mass — cart, the masses riding on it, and the hanging mass, all of it. The whole system accelerates together, so the whole system's mass resists. This is the single most common place to go wrong in this lab.

Now rearrange it the way the note does, into the form of a straight line:

$$a = (1 / M_T) \times F_{g,H}$$

This is y = mx. Put a on the vertical axis and $F_{g,H}$ on the horizontal axis, keep M_T fixed across every run, and the data should fall on a straight line through the origin whose slope is $1/M_T$. The slope has units of $(m/s^2)/N = 1/kg$, so one divided by the slope is a mass in kilograms.

◎ The two things you must confirm

"Verifying Newton's second law" is not a vague aim. In this lab it means two specific, checkable claims, and your Discussion has to address both by name:

① The relationship is linear
Acceleration rises in proportion to the driving force. Not merely "increases" — *in proportion*, a straight line, and one consistent with passing through the origin.

② The slope is $1 / M_T$
The number the graph gives you is the total mass of the system, and it agrees with the mass you get by weighing the components.

A straight line alone does not verify the second law — plenty of wrong theories predict straight lines. It is the slope *matching a mass measured independently* that turns a graph into a verification.

Predict first. The hanger carries 100 g and the system accelerates at some rate. You now take a further 100 g *off the cart* and put it on the hanger, doubling the hanging mass. What happens to the acceleration?

(A) It doubles.

(B) It roughly doubles, but falls a little short.

(C) It increases by less than double, because the hanging mass is in the denominator too.

(D) It stays the same — no mass entered or left the system.

Why that question is the whole design of this lab. Had you *added* 100 g from outside instead of transferring it, the acceleration would not have doubled: the extra mass would appear in the numerator *and* in the denominator, and you would be changing two things at once with no way to tell which caused what. Transferring mass changes the driving force while leaving M_T alone, which is the only reason the slope of your graph means anything. Your Experimental section must say this in your own words.

The apparatus

The equipment is from Vernier. A low-friction cart runs on a level track about 2.0 m long. An ultrasonic motion sensor at one end records the cart's position many times a second. At the far end the string passes over a pulley at the table edge and down to a hanger. The hanging mass falls, the string pulls the cart, and the two move together.

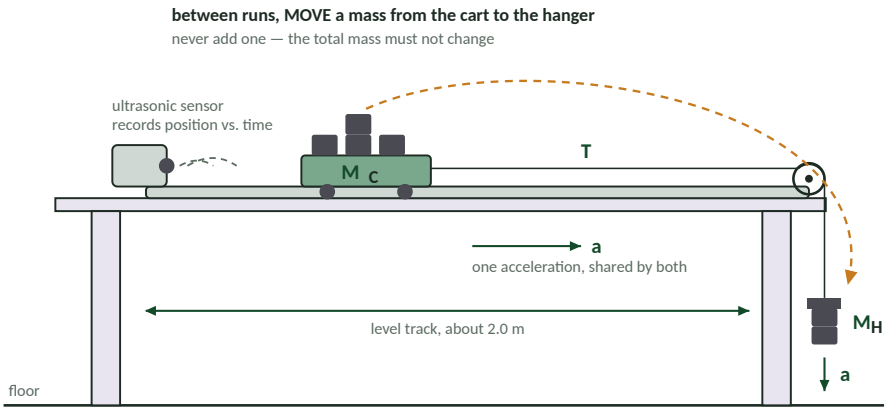


Figure A. The half Atwood machine. One light string runs from the cart over the pulley at the table edge and carries a single tension **T** throughout. The cart and everything on it, plus the hanger and everything on it, form one system of total mass M_T moving with one acceleration **a** — which is why the same letter labels the arrow on the track and the arrow below the hanger. The orange path is the procedure that makes the experiment work: mass is moved from cart to hanger between runs, so the driving force $F_{g,H} = M_H g$ rises while M_T stays exactly the same.

> **This is a virtual lab.** The five runs were recorded for you and are provided as Vernier Graphical Analysis files, and as a spreadsheet of the same position-and-time numbers if you would rather work there. You are still responsible for the analysis, the graph and the uncertainty — the measurement is what has been done for you, not the physics.

The masses

Weighed on the laboratory scale:

The components of the system. These three numbers, added, are your accepted mass.

Component	Mass
Cart	510.72 g
Block mass carried on the cart	537.30 g
Hanging masses, all five together	500.00 g
Total system mass M_T (accepted)	**1548.02 g = 1.54802 kg**

Every one of the five hanging masses starts out on the cart. Whichever ones are hanging at any moment, the sum above is unchanged — which is exactly the point.

Part I — The data

Five runs were recorded, with 100, 200, 300, 400 and 500 g on the hanger. In each run the cart begins at rest near the sensor, is released, accelerates along the track, and finally stops when the hanger reaches the floor or the cart reaches the end of the track.

1. Download the data (the cards at the foot of this page). Open a run in Graphical Analysis, or open the workbook if you prefer to work in a spreadsheet — they contain the same numbers.

2. For each run, compute the **driving force** $F_{g,H} = M_H g$, in newtons, from the hanging mass in kilograms. Do this by hand once, in full, with units, as your sample calculation.
3. Plot position against time and look at the shape before you fit anything. You should be able to see the three phases with your eye: a flat stretch before release, a curved accelerating stretch, and whatever happens after the motion ends.

Part II — Getting an acceleration out of each run

Nothing in this experiment measures acceleration directly. What you have is position against time, and for motion from rest under a constant acceleration, that is a parabola:

$$x(t) = x_0 + v_0\,t + \tfrac{1}{2}\,a\,t^2$$

Where this holds: constant acceleration only — which is why the fit window matters so much. Fit a parabola to a stretch of data where the acceleration was *not* constant and the number you get is not an acceleration, it is an average of several things.

Fit a quadratic to each run. The coefficient of t^2 in that fit is $\tfrac{1}{2}a$, so:

$$a = 2 \times (\text{quadratic coefficient})$$

Fit only the part of the run where the system is accelerating. Do not include the flat stretch before release, and do not include anything after the hanger lands or the cart reaches the end — the sensor produces genuine nonsense there, including dropouts and reflections. Say in your report what time window you used for each run and how you chose it. This is not a formality: refitting these runs over slightly different windows shifts the individual accelerations by a few percent, and an examiner cannot check your work without knowing which points you kept.

Record for each run: the hanging mass, the driving force, the fit window you used, the quadratic coefficient, the acceleration, and the uncertainty the fit reports for that coefficient.

Part III — The graph that weighs the system

Plot your five accelerations on the vertical axis against the five driving forces on the horizontal axis, and fit a straight line. From the rearranged equation, the slope of that line is $1/M_T$.

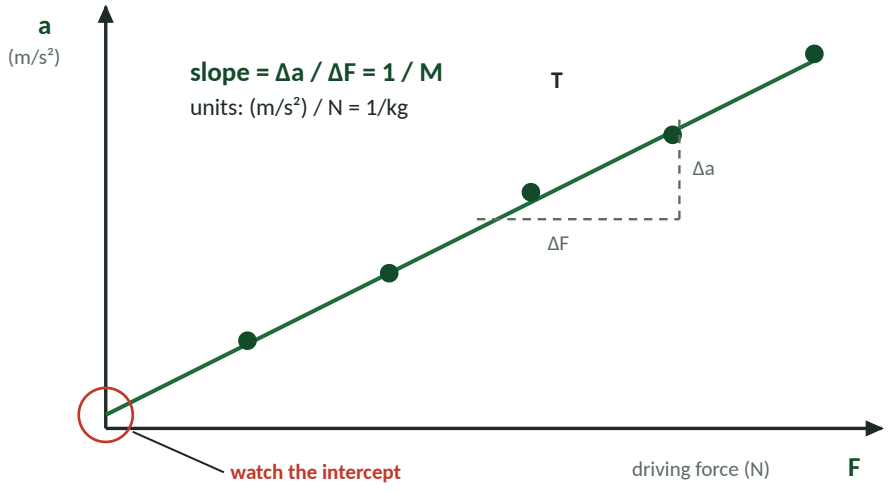


Figure B. What the analysis is doing. The slope carries the physics; one divided by it is the total mass of everything that moved. The intercept ought to be zero — no force, no acceleration — so if yours is not, that is a result in its own right and not a rounding error.

Getting the mass, and its uncertainty, from the slope

$$M_T = 1 / \text{slope}$$

Check the units as you go. A slope in 1/kg inverts to a mass in kg. If your slope came out in some other combination of units, you have mixed grams with kilograms somewhere, and the mass you report will be wrong by a factor of a thousand.

Your fitting tool reports an uncertainty on the slope. Because the mass is one over the slope, that uncertainty propagates as:

$$\delta M_T = \delta(\text{slope}) / (\text{slope})^2$$

Equivalently, the *fractional* uncertainty carries straight across: $\delta M / M = \delta(\text{slope})/\text{slope}$. Quote your experimental mass as $M \pm \delta M$, with both numbers to a sensible precision.

The comparison

Compare the mass from your graph with the accepted mass of 1548.02 g:

$$\%\ \text{error} = |M_{\text{experimental}} - M_{\text{accepted}}| / M_{\text{accepted}} \times 100\%$$

Then ask the question the percentage on its own cannot answer: is your experimental mass *higher* or *lower* than the accepted mass, and is that the direction you would expect? Work out what a real half Atwood machine does that the ideal one in Note 6 does not, and decide which way each of those effects would push your result. A percentage with a sign attached and a mechanism behind it is worth far more than a small percentage on its own.

Check your analysis

Paste your five force-and-acceleration pairs below to check the line you fitted. Your report still has to show the graph and a sample calculation worked by hand, with units — this tool checks arithmetic, it does not replace the analysis.

TOOL · FIT A AGAINST F, AND READ OFF THE MASS

One run per line: driving force in N, then acceleration in m/s², separated by a comma, a tab or a space

Accepted total mass (g)

1548.02

Fit

With an intercept (recommended)

Uses $g = 9.8$ m/s² where a force is computed from a mass. Uncertainties are the standard least-squares errors on the fitted parameters.

Additional Questions

All three are required, and all three are calculus questions — they ask you to derive things this handout has so far simply handed you. Answer them in a section of their own headed **Additional Questions**, in **sentences and paragraphs** with the working shown, never as a bare numbered list and never by restating the question before answering it. This section is worth 5 points on the rubric.

AQ1 — where the uncertainty formula comes from. You obtain the total mass from the slope as $M_T = 1/s$, and Part III handed you $\delta M_T = \delta s / s^2$ without justifying it. Derive it. M_T depends on one variable, so a small error in the slope produces $\delta M_T \approx |dM_T/ds|\ \delta s$ — differentiate $M_T = s^{-1}$ and show the formula follows. Then show that the *fractional* uncertainty is unchanged by

the inversion, $\delta M/M = \delta s/s$, and state what the same argument gives for any quantity depending on the slope as s^n. Finish by applying it to your own fitted slope and quoting $M_T \pm \delta M_T$.

AQ2 — why the quadratic coefficient is half the acceleration. You have been using $a = 2 \times$ (quadratic coefficient) as a recipe. Justify it: differentiate $x(t) = x_0 + v_0 t + \tfrac{1}{2}at^2$ twice and say what each coefficient of the fit physically is. State precisely what that rule assumes about d^2x/dt^2, and use that to explain what actually defines a correct fit window — not "the part that looks curved," but a condition on the second derivative. Then differentiate your own fitted parabola once to get $v(t)$, evaluate it at both ends of your window, and compare those two numbers against velocities taken straight from the raw data by central differences, $v(t_i) \approx (x_{i+1} - x_{i-1}) / 2\Delta t$. They should agree. Say which of the two you would trust more, and why, given that the sensor's position noise is a few millimetres.

AQ3 — the experiment you were forbidden to do. Because mass is transferred, the total stays fixed and $a = (g/M_T)M_H$ exactly; $da/dM_H = g/M_T$ is a constant, which is the real reason your graph is a straight line. Now work out what the other experiment would have produced. Had mass been *added* from outside, the cart and its load would have stayed at $M_C = 1048.02$ g while the hanger grew, giving $a(M_H) = M_H g / (M_C + M_H)$. Expand this about $M_H = 0$ to second order; show the leading term is a straight line of slope g/M_C and that the first correction bends it downward. Compute how far below that initial tangent the true curve falls at each of the five hanging masses used here, and find the hanging mass at which the deviation first reaches 10%. Then answer the question that matters: given the scatter you actually see in your own five accelerations, would that curvature have been visible over the range 100–500 g — or would the wrong experiment have passed for a straight line?

Prompts for your Discussion

These are **not** the Additional Questions. Work them into your **Discussion** section as part of interpreting your own results, in sentences and paragraphs. Do not limit yourself to these; add any other insight you have.

I. Look at your intercept. In an ideal experiment it is zero, because zero force should produce zero acceleration. Is yours zero within its uncertainty? If it is not, work out what constant force would account for it, and say whether the sign of that force makes physical sense for this apparatus.

II. What factors would make the mass from your graph differ from the mass on the scale? For each one you name, say which *direction* it pushes the result, and estimate its size if you can. At least one thing in this apparatus has mass, is being accelerated, and never appears anywhere in your total — find it.

III. Refit one of your runs using a slightly different time window — a few points more at one end, a few fewer at the other. How much does the acceleration change? What does that tell you about the precision you are entitled to claim?

> **One thing the rubric will not accept:** "human error." It explains nothing, and in a virtual lab where the data were collected by a machine it explains even less. If your percent error is under 10%, attribute it to ordinary experimental uncertainty and say so explicitly, backed by the uncertainty on your slope. If it is over 10%, you must identify specific error sources and *quantify* them — a named mechanism with a number attached.

★ Optional bonus — up to +3 points

Harder than anything else on this page. Skipping it costs you nothing.

Your line is straight, your fit is good, and your mass is still several percent off. Two quite different mechanisms produce exactly that symptom, and this problem asks you to tell them apart.

I. Suppose the only flaw is that something in the apparatus has mass, is accelerated, and is missing from your M_T — the pulley and the string. Show that this multiplies every acceleration by one constant factor, so it changes the *slope* of your graph while leaving the motion exactly parabolic.

II. Now suppose instead there is a drag force proportional to speed, so the system obeys $M\,dv/dt = F - bv$ with $v(0) = 0$. Solve that differential equation for $v(t)$, and integrate once more for $x(t)$. Expand $v(t)$ for small bt/M and show that a parabola fitted over a window of mean time $\bar{t}$ returns not the true $a = F/M$ but approximately $a(1 - b\,\bar{t}/M)$.

III. Here is what makes it awkward: that bias is still proportional to F. On an a-against-F graph, drag therefore *counterfeits* an inertia excess of roughly $\Delta m \approx b\,\bar{t}$. Estimate the value of b that would account for your own percent error.

IV. Design the measurement that separates the two. One of them predicts something the other does not, and you already have the data to look for it — the third Discussion prompt above is the hint.

End by saying plainly whether your own data settle the question or are simply too noisy to. **Either answer earns full credit if it is argued from the numbers.**

✓ Before you submit

Every line here maps to a row on the Lab 04 grading rubric.

FORMAT & FRONT MATTER

- [ ] Lab Template used, with names, title, and date in the heading
- [ ] Abstract: four sentences or fewer — theme, the experimental mass with its uncertainty, the percent error. Nothing else.
- [ ] Introduction carries *your own* derivation of $a = F_{g,H}/M_T$ from the two equations, every symbol defined, and a hypothesis you can later confirm or reject
- [ ] The rearrangement to $a = (1/M_T)F$ is shown, and the report says in words why that makes the slope a mass

EXPERIMENTAL

- [ ] Labelled sketch of the half Atwood setup, redrawn by you — not this page's figure pasted in
- [ ] Free-body diagram for *each* body, with the sign convention chosen for each stated explicitly
- [ ] Procedure in paragraph form, in your own words, including how the position-time data were collected and converted into an acceleration
- [ ] **The constant-total-mass strategy is explained** — that mass was moved from cart to hanger rather than added, and why that is what makes force the single variable

RESULTS

- [ ] Table of the five runs: hanging mass, driving force, fit window, quadratic coefficient, acceleration, and the fit's uncertainty
- [ ] At least one position-versus-time graph shown with its fitted parabola, so the reader can see which points were used
- [ ] The acceleration-versus-force graph, with the fitted line, axes labelled with units, numbered as a Figure with a caption and described in the text
- [ ] Every table numbered, captioned, with units and consistent significant digits, referenced in the text ("Table 1 shows…")
- [ ] Experimental mass quoted as $M \pm \delta M$, with the propagation from the slope's uncertainty shown
- [ ] A sample calculation, worked step by step with units, for *each* type: driving force, acceleration from the fit, mass from the slope, uncertainty propagation, percent error

DISCUSSION & CONCLUSION

- [ ] Discussion opens with "we experimentally determined that… ± error"

- [ ] **Both** verification criteria addressed by name: that the relationship is linear, *and* that the slope equals $1/M_T$
- [ ] The intercept is discussed, not ignored
- [ ] The direction of the discrepancy is explained, not just its size
- [ ] States whether your introduction's hypothesis is confirmed, and argues confidence using the 10% rule
- [ ] All three Additional Questions (AQ1–AQ3) answered in a dedicated section, in paragraphs, with the calculus worked and shown
- [ ] The phrase "human error" appears nowhere in the report
- [ ] Conclusion: 3–4 sentences — the numerical result with percent error, a definitive verified / not-verified statement, and one concrete way to improve the measurement

REFERENCES

- [ ] One to three references beyond the textbook — this handout, Note 6, and an outside source on Newton's second law or on the analysis software
- [ ] Author, title, and date given for each — a bare URL is not a citation

0 of 24 checked.

Files for this lab

HANDOUT

This lab, as a PDF

The whole page, printable. The calculator and the checklist only work on the web version.

REQUIRED

The five runs (Graphical Analysis)

100 g through 500 g hanging, as .gambl files. Open each in Vernier Graphical Analysis and fit the accelerating part of the curve.

REQUIRED

Data and analysis workbook (XLSX)

The same five runs as position-and-time columns, plus the analysis layout with live formulas, and a worked example sheet so you can see the shape of a finished analysis.

REQUIRED

Lab 04 grading rubric

The rubric page, with Word and PDF copies. What each section is worth. Every line of the submission checklist above maps to a row here.

REQUIRED

Report starter (DOCX)

The Lab Template with this lab's headings in place. Lighter than Lab 3's — see the note inside it.

START HERE

M3 N6 — Verifying Newton's Second Law with a Half Atwood Machine

The note this lab measures. Its derivation and diagrams belong in your Introduction and Experimental sections.

REFERENCE

Formatting Data Tables for Lab Reports

Table numbering, captions, significant digits, units.

Abstract rules, section-by-section guidance and the submission window live in the **Writing Lab Reports** module on Brightspace, alongside the Module 3 Lab folder where you hand this in.

References to get you started

Your references should include material on Newton's second law and on the software used to analyse the motion. Your textbook, this handout and the Module 3 note pages all count, and you need at least one source beyond the textbook. The note this lab measures is Module 3 Note 6.

1. [Author's names], *Journal or Book title*, edition, Volume number, page number, (year).

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